TEXAS A&M UNIVERSITY

Functions

CSCE 222: Discrete Structures for Computing

Outline

- Relations
- Functions

Functions

A function f from set A to set B is an assignment of exactly one element of B to each element of A.

$f: A \rightarrow B$ and $f(a) = b$.

Ways to specify functions:

- give a formula, e.g. $f(x) = x + 1$
- give an algorithm to compute the function

Every function is a relation but not vice versa!

Functions

Given sets A and B.

F is a function from A to B for any $x \in A$ where $y \in B$ such that $(x,y) \in F$, we use the notation $y = F(x)$

Activity

A = {3,5,7} and B = {2,4}

Assume that we have the following relation

R = {(3,2), (3,4), (7,2)}

Is the relation R also a function F from A to B?

Activity

A = {3,5,7} and B = {2,4}

Assume that we have the following relation

S = {(3,2), (5,4), (7,2)}

Is the relation S also a function F from A to B?

Function Important Terminologies

For a function $f: A \rightarrow B$ with $f(a) = b$

A is the domain

B is the codomain

b is the image of a

a is a preimage of b (preimage is not necessarily unique)

B is the range of f (images of all elements of A)

Function Terminologies

Suppose $f: Z \rightarrow Z$ is $f(x) = x^2$

Domain is:

Codomain is:

Image of 3 is:

Preimages of 16:

Range is the set of all integers that are perfect squares

Function Terminologies

Suppose $f: Z \rightarrow Z$ is $f(x) = x^2$

Domain is Z (set of integers)

Codomain is Z

Image of 3 is 9

Preimages of 16 are 4 and -4

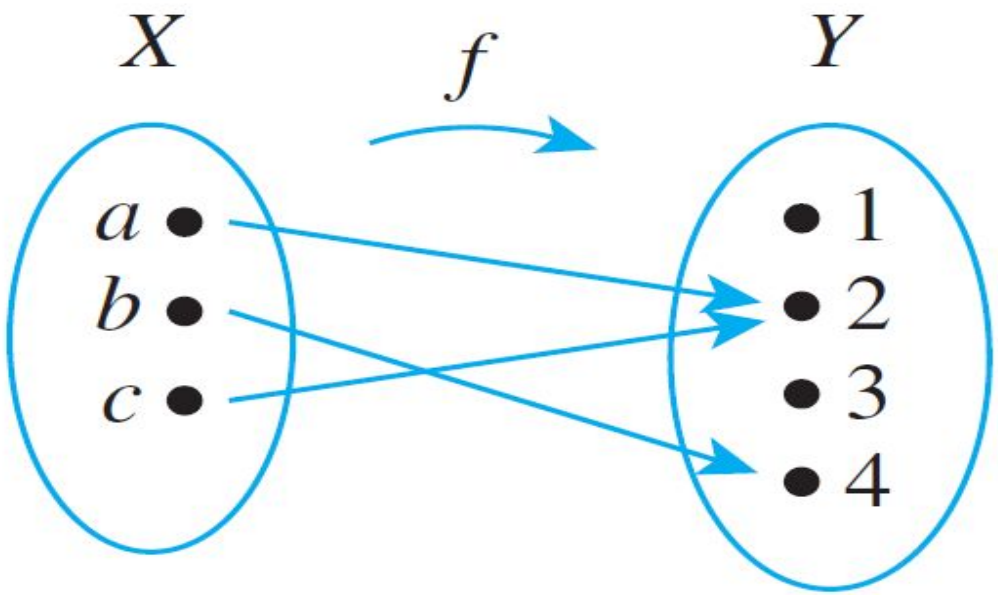
Range is the set of all integers that are perfect squares

Arrow Diagrams

An arrow diagram defines a **function** if the diagram has the following properties:

1. Every element of X has an arrow coming out of it.
2. No element of X has two arrows to two different elements of Y.

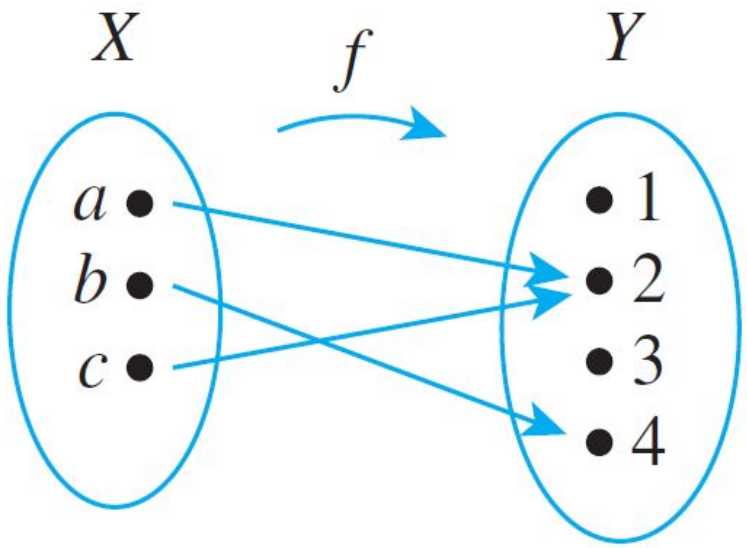
Function



X = {a, b, c}, Y = {1, 2, 3, 4}

We define a function f from X to Y as follow:

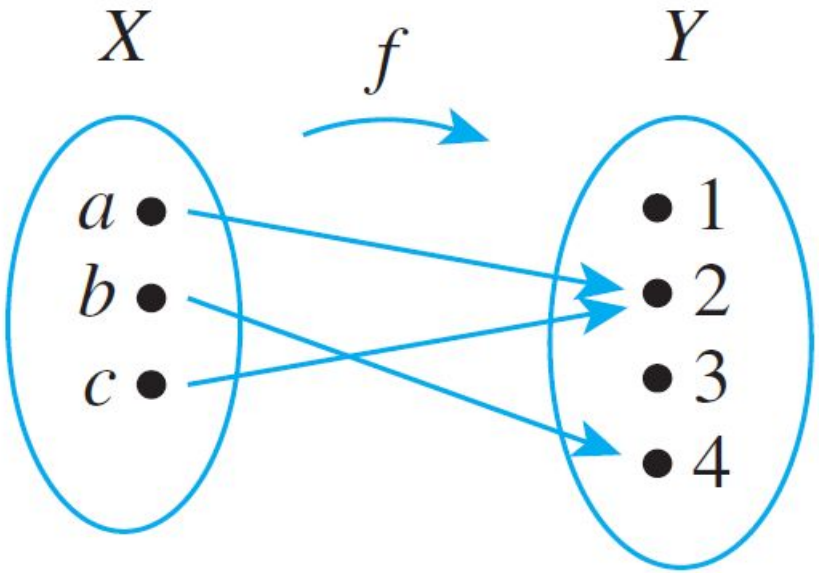
Domain and co-domain



The domain X of f is {a, b, c}

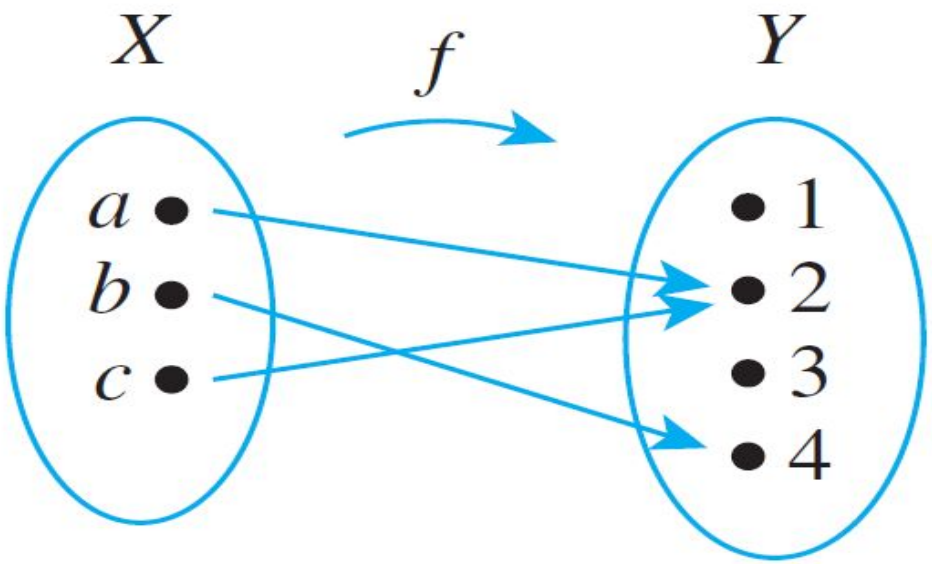
The co-domain Y of f is {1, 2, 3, 4}

Find f(a), f(b), and f(c)



f(a)=2, f(b)=4, and f(c)=2

Range



The range of f is defined as {y ∈ Y | y = f(x), for x in X}

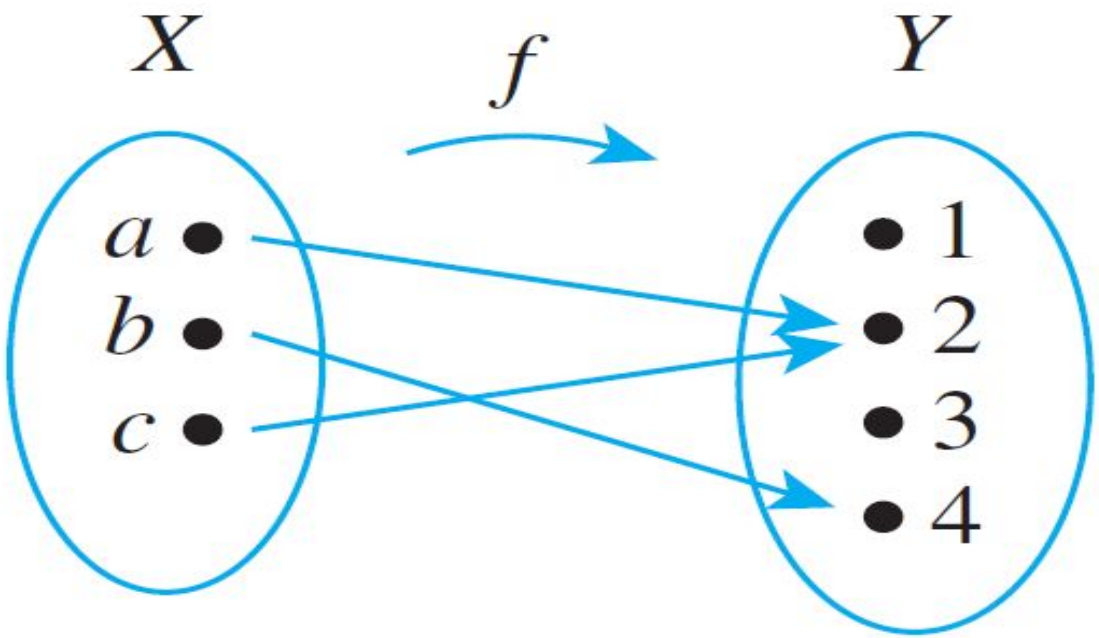
i.e., what values of Y are mapped to by X?

The range is {2,4}

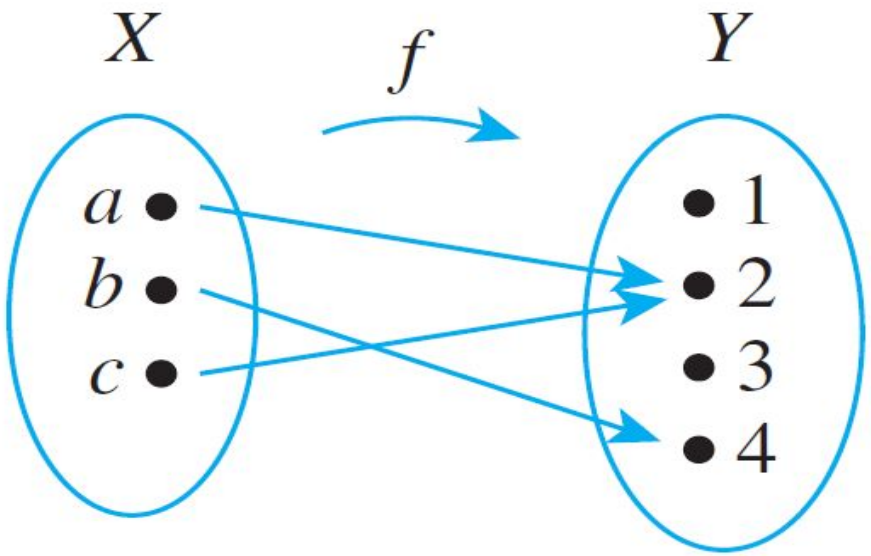
Preimage

Is c a preimage of 2?
Yes, since f(c) = 2

Is b a preimage of 3?
No, since f(b) ≠ 3



Preimage



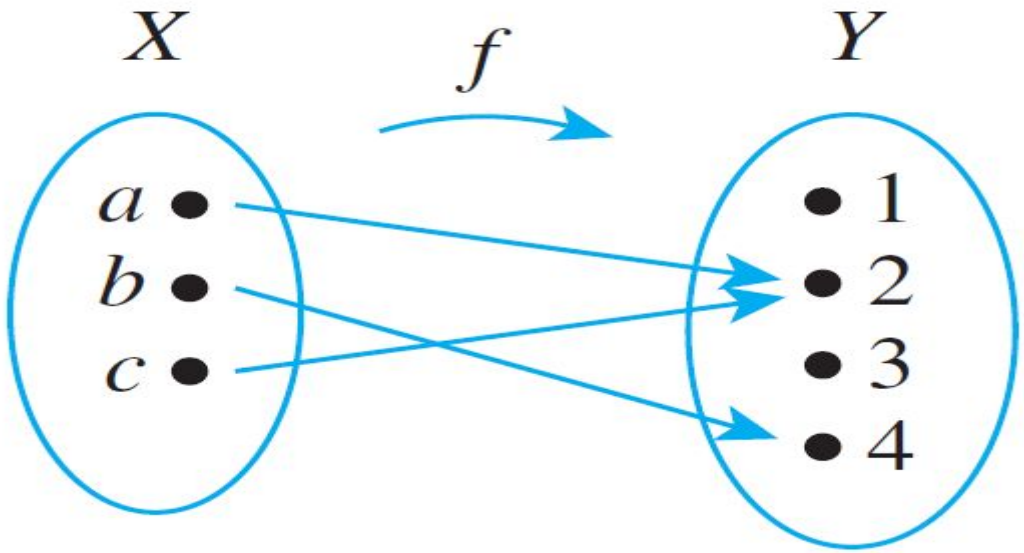
Find the preimages of 2, 4, and 1.

Both f(a) and f(c) = 2, so the preimage of 2 is {a,c}

The preimage of 4 is {b}

There is no x ∈ X where f(x) = 1, so the preimage of 1 is ∅

Ordered pairs



Represent *f* as a set of ordered pairs.

Ordered pairs are (x, f(x)):

(a,2), (b,4), (c,2)

Function Equality

Given functions F: $X \rightarrow Y$ and G: $X \rightarrow Y$

Then: $F = G \leftrightarrow F(x) = G(x) \ \forall \ x \in X$

Activity

Assume we have a set A = {0, 1, 2}

Also assume we have the following functions defined on set A:

$$f : A \rightarrow A$$

$$\forall x \in A, f(x) = \left(x^2 + x + 1\right) \bmod 3$$

$$g : A \rightarrow A$$

$$\forall x \in A, g(x) = (x+2)^2 \bmod 3$$

Are functions f(x) and g(x) equal?

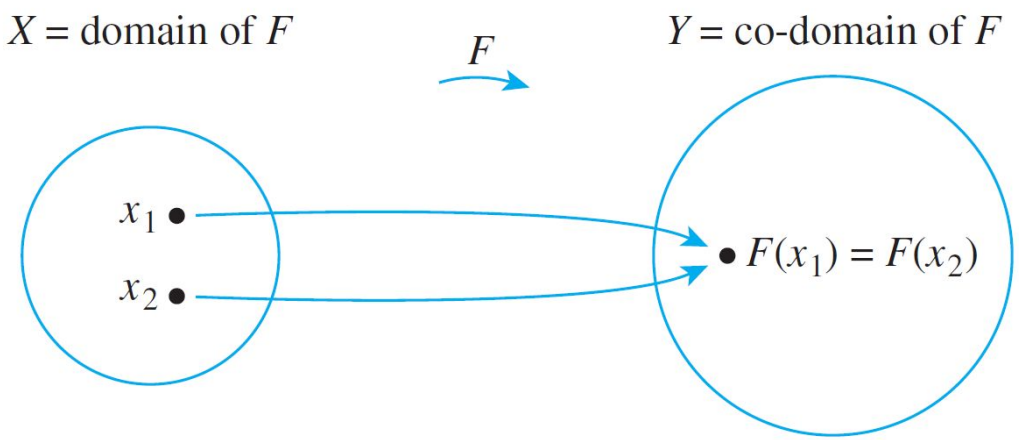
Classes of functions

- One-to-One Functions
- Onto Functions
- One-to-One Correspondence
- Inverse Functions

One-to-One Functions

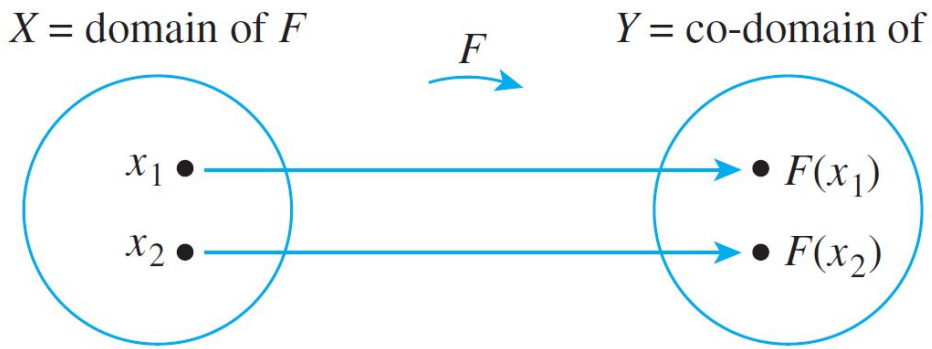
Suppose we have the following two functions

(1)



Two distinct elements of X are sent to the same element of Y.

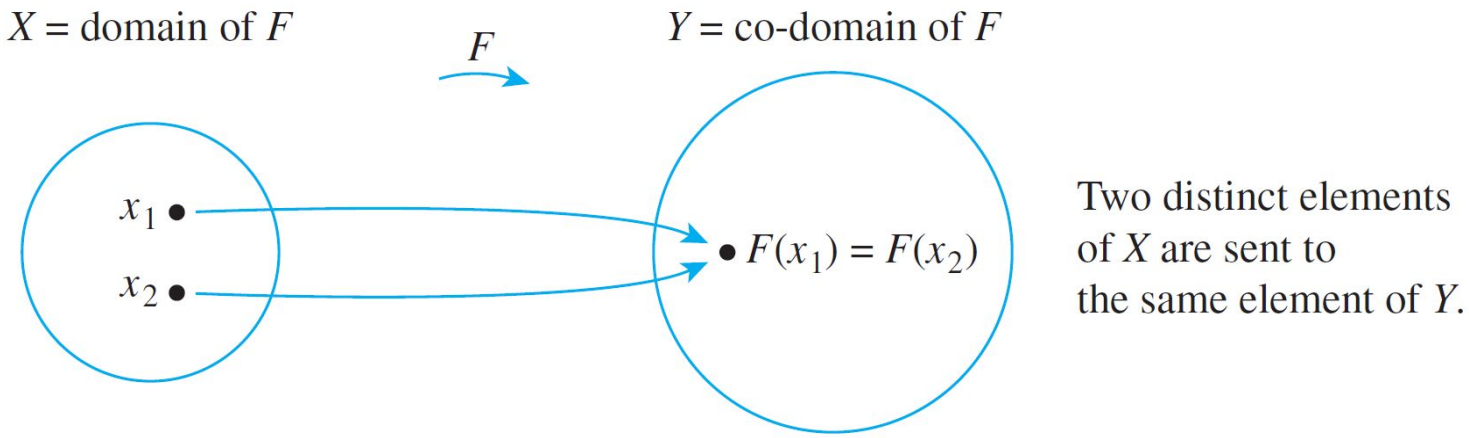
(2)



Any two distinct elements of X are sent to two distinct elements of Y.

One-to-One Functions

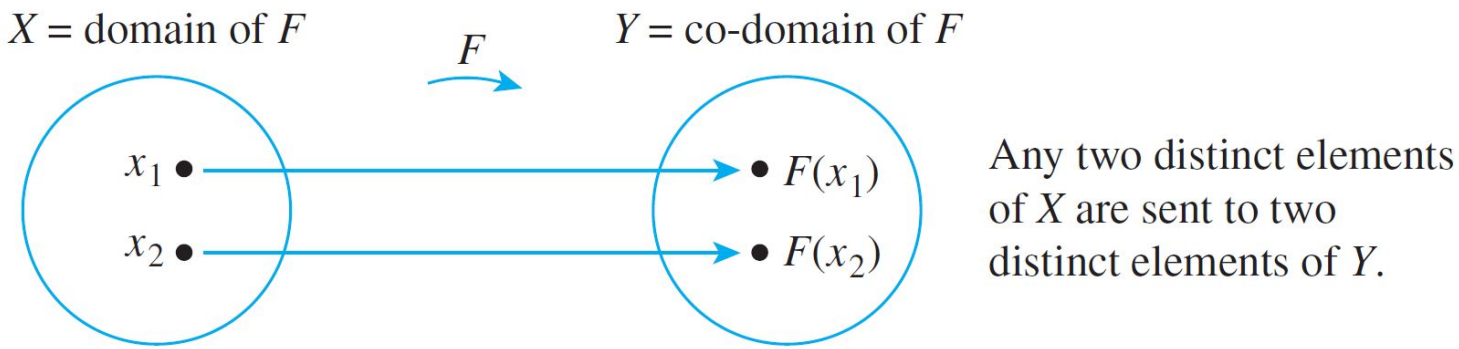
(1)



For function (1), each x maps to the same y: $F(x_1) = F(x_2)$

One-to-One Functions

(2)



For function (2), each x maps to a unique y: $F(x_1) \neq F(x_2)$

Function (2) is called a one-to-one function.

One-to-One Functions: Definition

Let F be a function from set X to set Y

F is one-to-one iff for all elements x1 and x2 in X:

- If $F(x_1) = F(x_2)$, then $x_1 = x_2$, or
- If $x_1 \neq x_2$, then $F(x_1) \neq F(x_2)$

Symbolically:

$$F : X \to Y \text{ is one-to-one} \Leftrightarrow \forall x_1, x_2 \in X, \left(F(x_1) = F(x_2)\right) \to \left(x_1 = x_2\right)$$

Also called an "injective function" or an "injection"

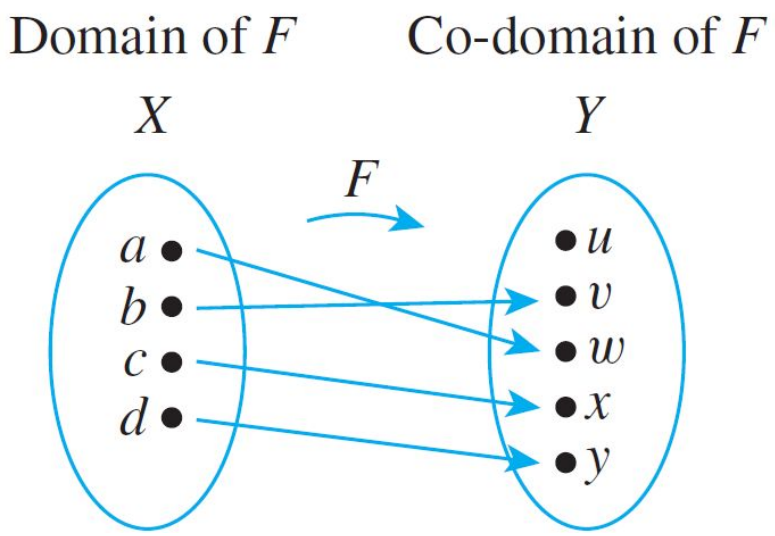
Functions that are not one-to-one

A function is not one-to-one if the negation of the previous statement is true:

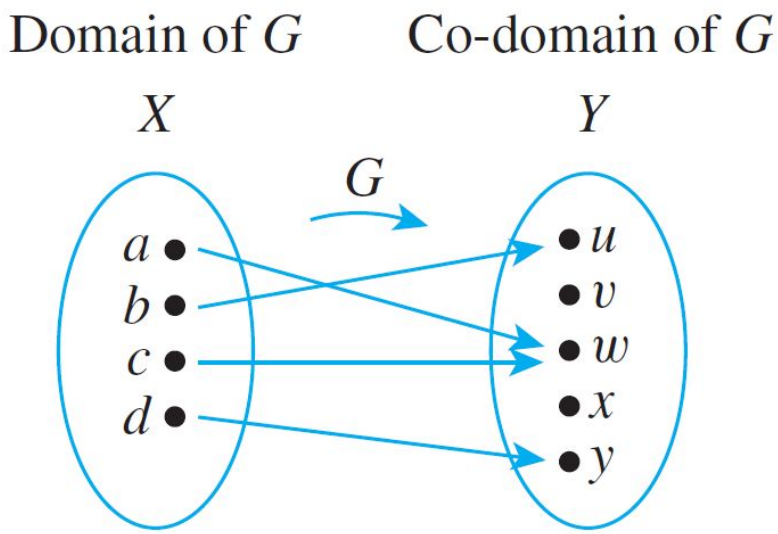
A function $F: X \to Y$ is *not* one-to-one $\Leftrightarrow$ $\exists$ elements x_1 and x_2 in X with $F(x_1) = F(x_2)$ and $x_1 \neq x_2$.

Activity

Are any of the following functions one-to-one?
Why or why not?



A

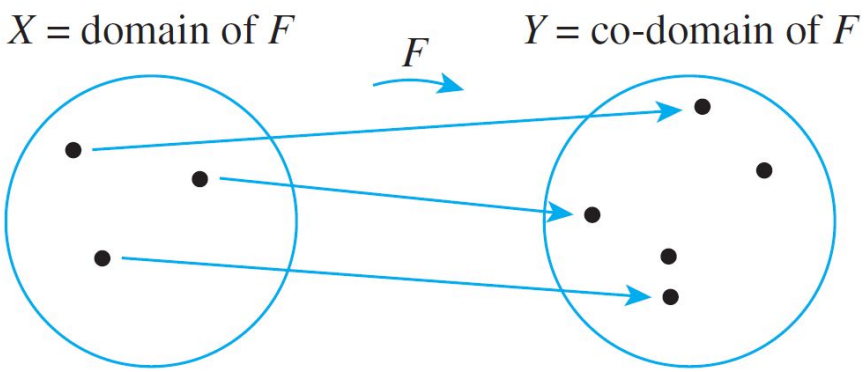


B

Onto Functions

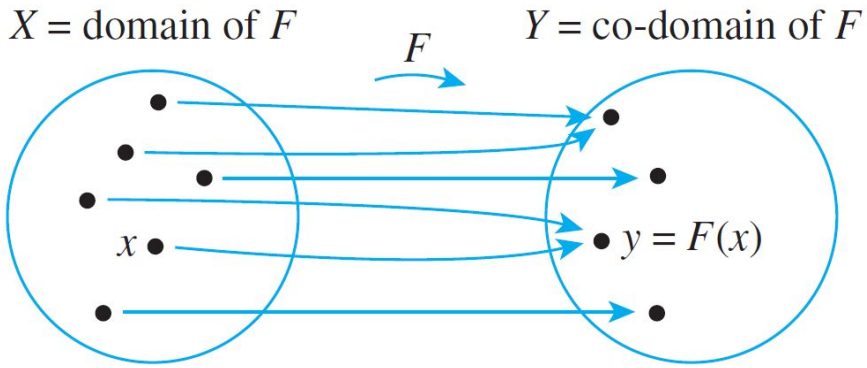
Suppose we have the following two functions

(1)



At least one element in Y does not equal $F(x)$ for any x in X.

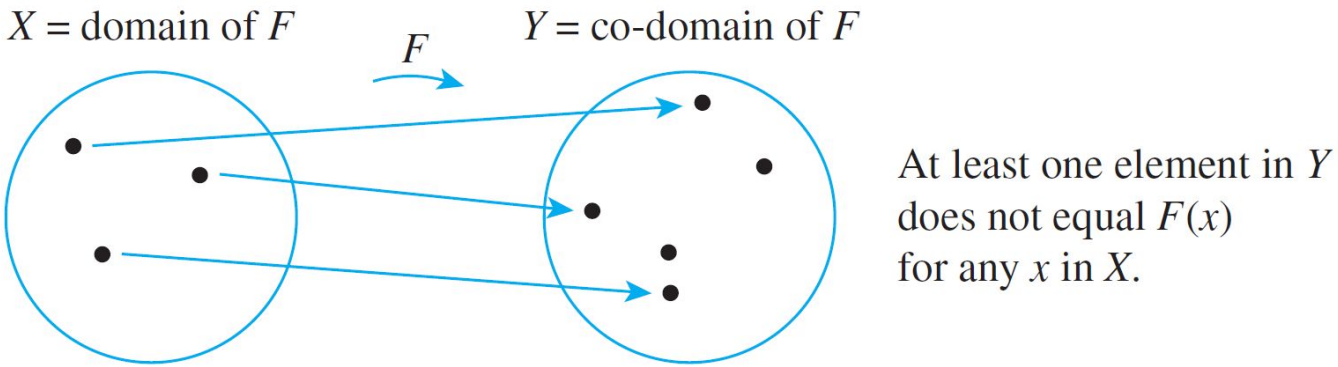
(2)



Each element y in Y equals $F(x)$ for at least one x in X.

Onto Functions

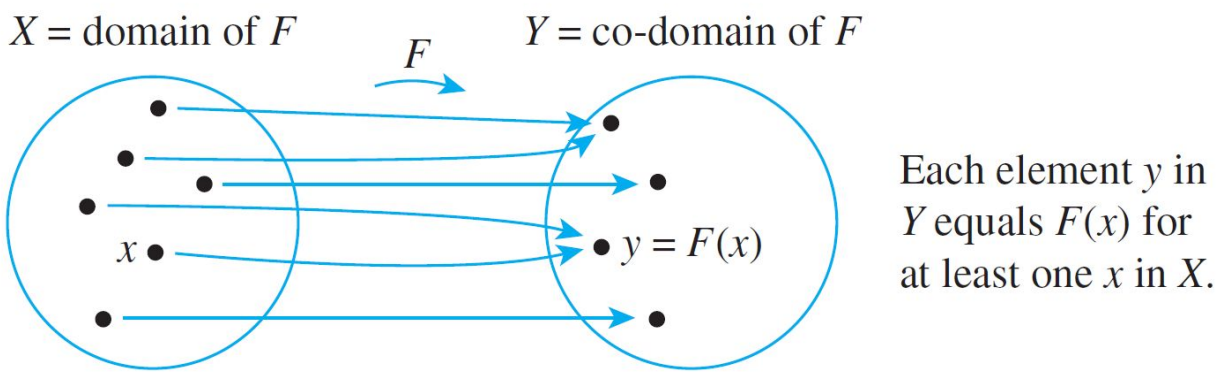
(1)



For function (1), there is at least one y that is not F(x) for any x

Onto Functions

(2)



For function (2), every y = F(x) for at least one x

Function (2) is called an onto function.

Onto Functions Definition

Let F be a function from set X to set Y

F is onto iff

- Given any y in Y, there is at least one x in X where y = F(x)

Symbolically:

$$F : X \to Y \text{ is onto} \Leftrightarrow \forall y \in Y, \exists x \in X \mid F(x) = y$$

Also called a surjective function or a surjection

Functions that are not onto

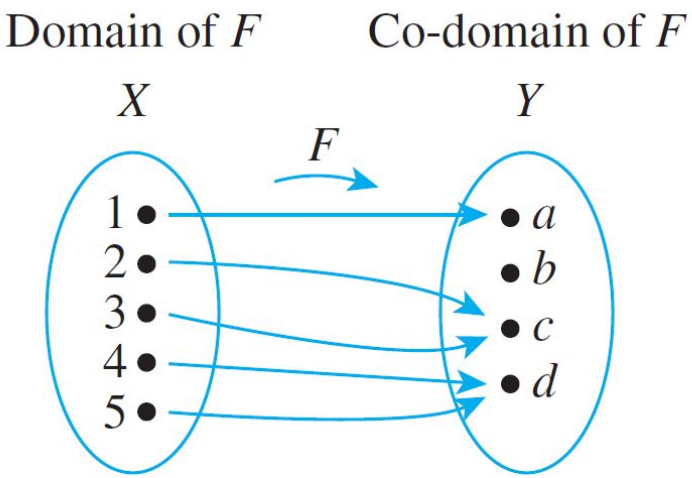
A function is not onto if the negation of the previous statement is true:

$$F\colon X \to Y \text{ is } \textit{not} \text{ onto} \quad \Leftrightarrow \quad \exists y \text{ in } Y \text{ such that } \forall x \in X,\ F(x) \neq y.$$

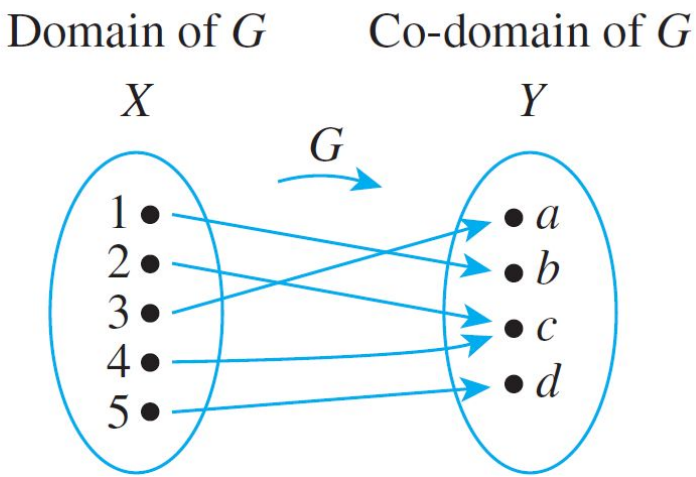
Activity

Are any of the following functions onto?

Why or why not?



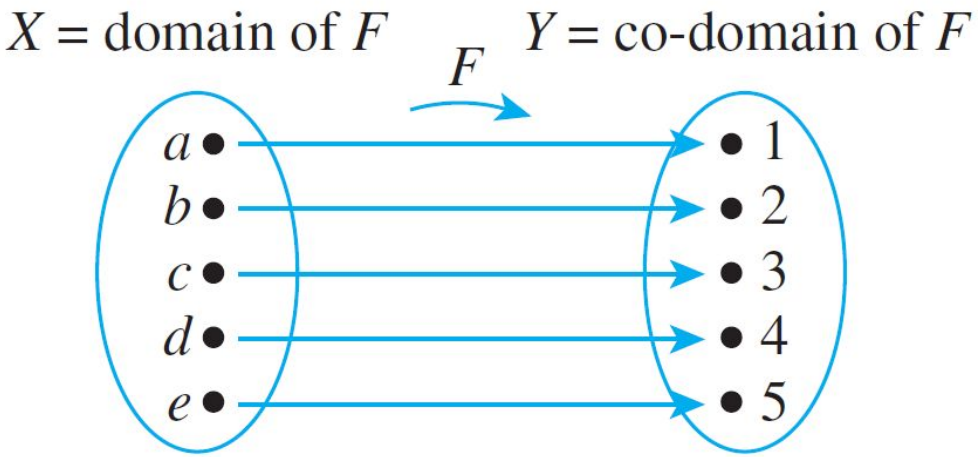
A



B

One-to-One Correspondence

Suppose we have the following function



Is F **one-to-one**? Yes, each F(x) has a distinct x

Is F **onto**? Yes, for all y, there is an x such that y = F(x)

Therefore, there is a **one-to-one correspondence** from X to Y

One-to-One Correspondence

Let F be a function from set X to set Y

F has one-to-one correspondence from set X to set Y iff F is one-to-one, and F is onto

F is also called a bijection.

Activity

$X = \{1,5,9\}$ and $Y = \{3,4,7\}$

Function $f: X \rightarrow Y$ is defined as: (1,4), (5,7), (9,4)

Are these functions one-to-one? Onto?

Activity

$X = \{1,5,9\}$ and $Y = \{3,4,7\}$

Function $g: X \rightarrow Y$ is defined as: $g(1) = 7$, $g(5) = 3$, $g(9) = 4$

Are these functions one-to-one? Onto?

Inverse Function

Suppose function F has a one-to-one correspondence from set X to set Y

Then there is a function from Y to X that "undoes" the action of F

This function is called the inverse function for F

Inverse Function Definition

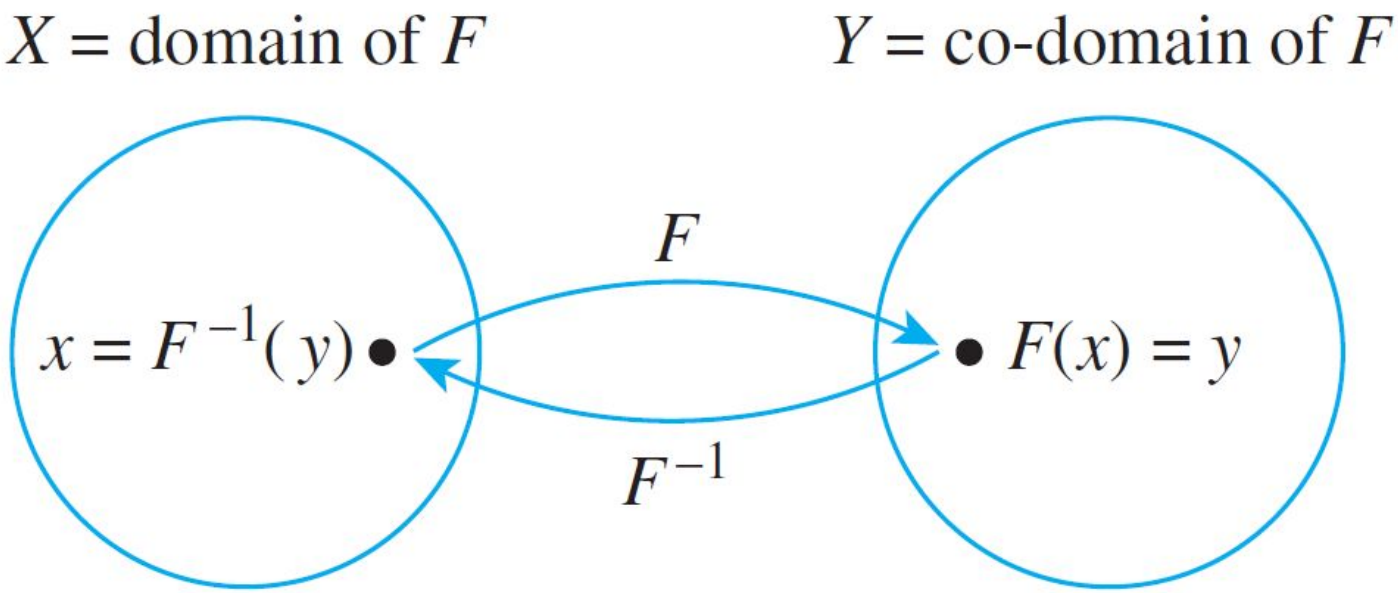
Given F: X→Y that is a one-to-one correspondence

There is a function F^{-1}: Y→X where

$$F^{-1}(y) = x \Leftrightarrow y = F(x)$$

$F^{-1}(y)$ is called the inverse function for F

Inverse Function Arrow Diagram



Inverse Function Example

$F(x) = 4x - 5$

1. Each x maps to a unique y, so F is one-to-one
2. Every $y = F(x)$ for some x, so F is onto

Thus, F has a one-to-one correspondence

Therefore, we can determine F^{-1}

Inverse Function Example

$$F(x) = y = 4x - 5$$

$$y + 5 = 4x$$

$$F^{-1}(y) = x = \frac{y+5}{4}$$

Floor and Ceiling

Floor and ceiling functions work on real numbers

For a real number x:

- The floor of x rounds x down to the nearest integer value
- The ceiling of x rounds x up to the nearest integer value

Floor and Ceiling

Ceiling(x) = 4

Round UP to nearest integer

x = 3.141592654

Round DOWN to nearest integer

Floor(x) = 3

Floor Definition

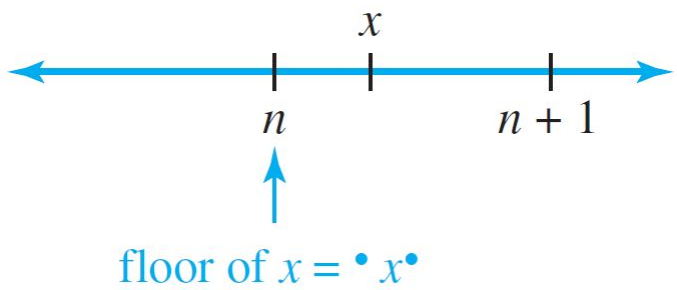
• Definition

Given any real number x, the **floor of x,** denoted $\lfloor x \rfloor$, is defined as follows:

$$\lfloor x \rfloor = \text{that unique integer } n \text{ such that } n \leq x < n + 1.$$

Symbolically, if x is a real number and n is an integer, then

$$\lfloor x \rfloor = n \quad \Leftrightarrow \quad n \leq x < n + 1.$$



Ceiling Definition

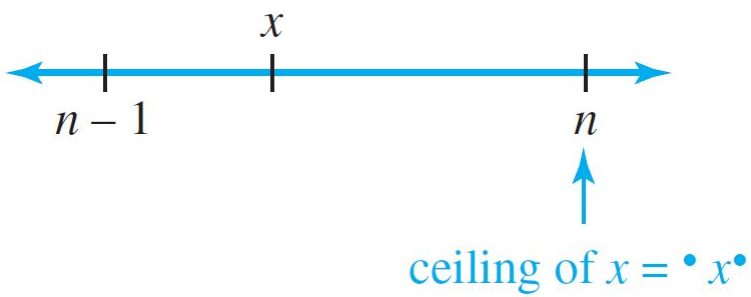
• Definition

Given any real number x, the **ceiling of x,** denoted $\lceil x \rceil$, is defined as follows:

$$\lceil x \rceil = \text{that unique integer } n \text{ such that } n - 1 < x \leq n.$$

Symbolically, if x is a real number and n is an integer, then

$$\lceil x \rceil = n \quad \Leftrightarrow \quad n - 1 < x \leq n.$$



Floor and Ceiling

Ceiling(x) = -3

Round UP to nearest integer

x = - 3.141592654

Round DOWN to nearest integer

Floor(x) = -4

Examples

1. How many **bytes** are needed to store 100 bits of data?

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$\lceil 100/8 \rceil = \lceil 12.5 \rceil = 13$ bytes

Examples

2. A cell is a unit of **53 bytes**. If the data rate is **500 kilobits** per second, how many **complete cells** can be transmitted in **1 minute**?

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In 1 minute, $500 \times 1000 \times 60 = 30$ Mbits can be transmitted.

Each cell is $53 \times 8 = 424$ bits.

$\lfloor 30M / 424 \rfloor = 70{,}754$ complete cells

Application to C/C++/Java

```
double x, x1, x2;

x1 = floor(x);    // floor of x
x2 = ceil(x);     // ceiling of x
```